



Mathematical foundations in data science

Inferential Statistics - 1

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The concept of inferential statistics



- Descriptive versus Inferential. The use of Sample and Population
 - Population as a bigger data set
 - Population as a phenomena (random variable vs variates)
- Some examples:
 - Marketing: Our discrete example of sales calls
 - Physical systems: Our continuous example of ball drop
 - Operations: The weight of bags of chips
 - Finance: Stock returns/price
- The two-sample (and multiple sample) setting

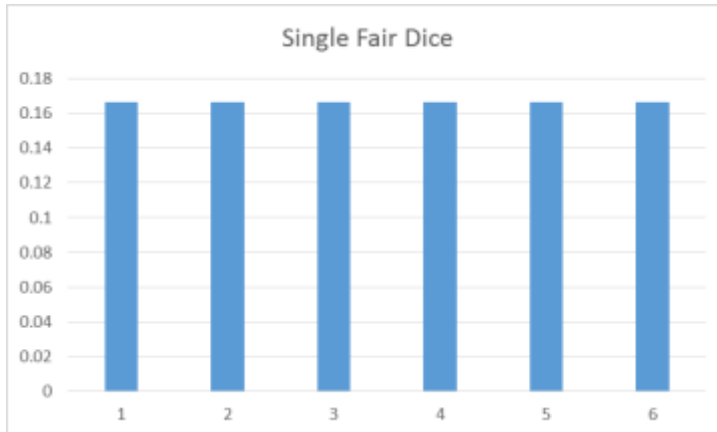


So where can I use it?

- Making an inference about a population from a sample
 - Given a sample statistic ($\bar{x}$) what can I say about the population parameter (μ):
Confidence Intervals
 - Given a sample can I answer pointed questions about the population: Hypothesis Tests

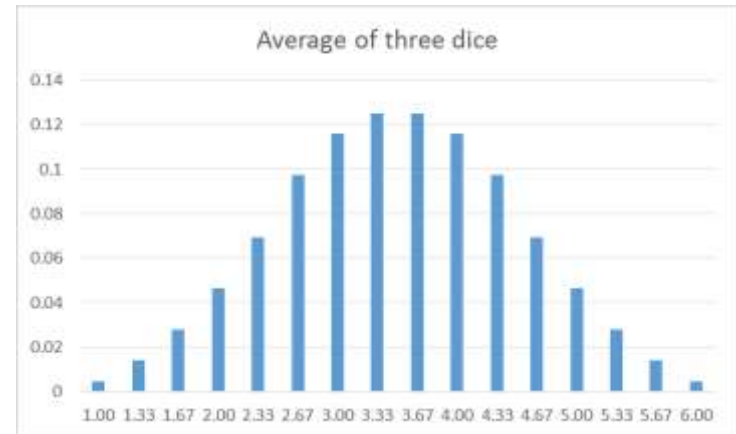
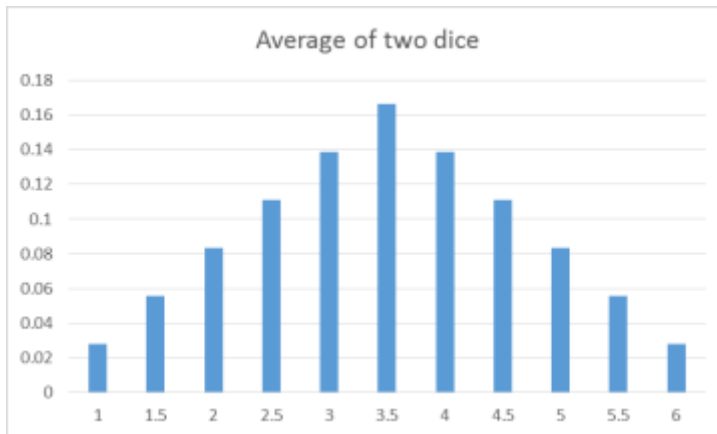
Central Limit Theorem

- The aggregation of a sufficiently large number of independent random variables results in a random variable which will be approximately normal.



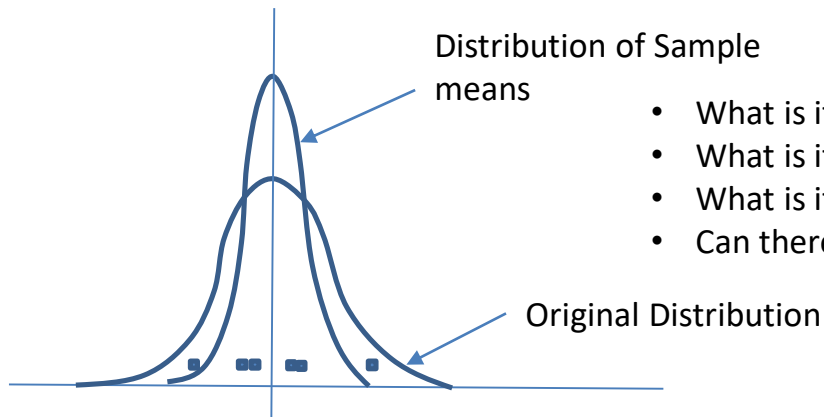
Central Limit Theorem

- More distributions:



Sampling distribution

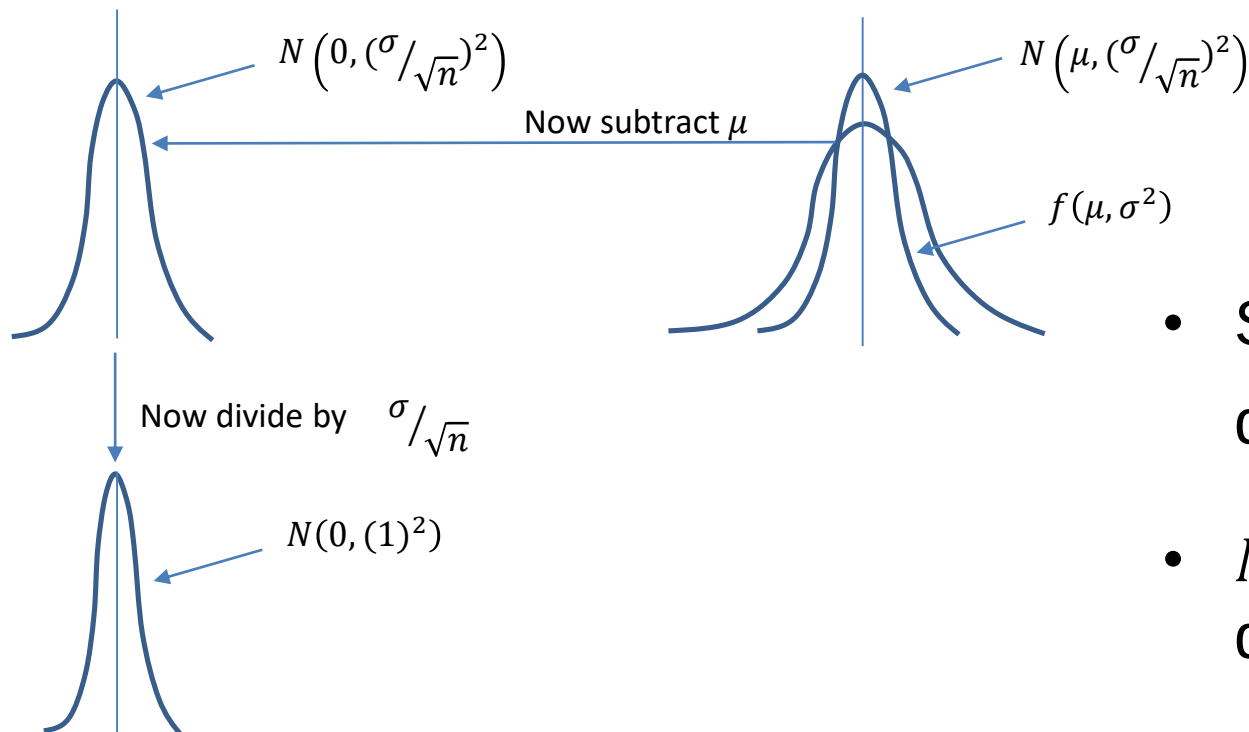
- Sampling distribution



- What is its shape?
- What is its mean?
- What is its standard deviation?
- Can there be a distribution for sample standard deviations?

Single sample interval

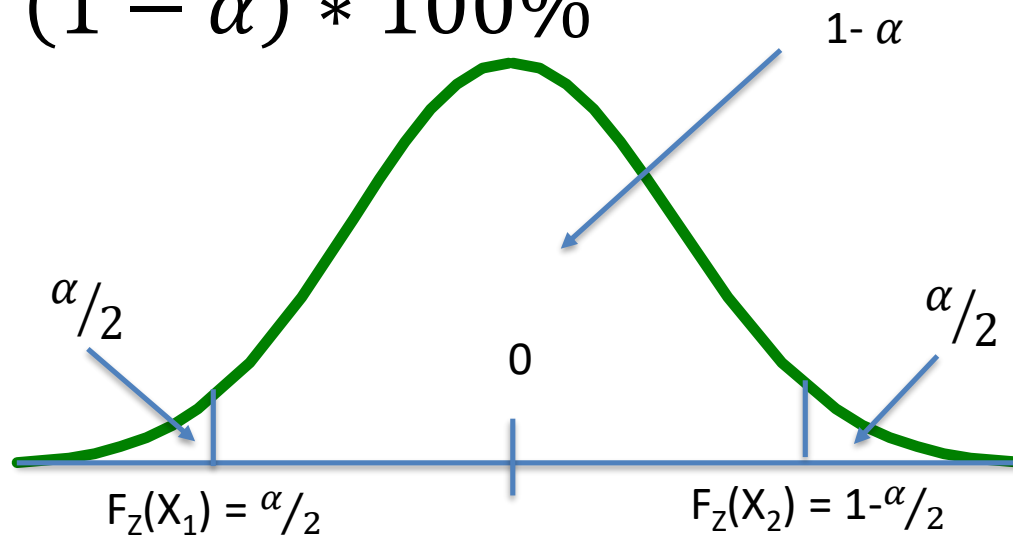
- Idea: if I can look at how much $\bar{X}$ can deviate from μ , then for a given $\bar{x}$, I can quantify the range within which μ could exist.



- So what is the distribution of $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$
- $N(0, 1^2)$ or Z distribution

Single sample interval for mean

- How far does the $N(0,1^2)$ or Z extend on either side?
- Certainty as percentage: a small chance of an error α , implies that we are certain with a $(1 - \alpha) * 100\%$



X_1 is referred to as $-z_{\alpha/2}$ and X_2 is referred to as $z_{\alpha/2}$

Steps

Therefore

$$P[-z_{\alpha/2} \leq Z \leq z_{\alpha/2}] = 1 - \alpha$$

$$P[-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}] = 1 - \alpha$$

$$-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}$$

$$-z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \leq \bar{X} - \mu \leq +z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$\bar{X} - z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{X} + z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$



Examples and discussion



- Examples
 - The ball drop example
 - What is the average waiting time for a patient in doctor's office?
 - What is the average diameter of a part we manufacture?
 - What is the average rating (on 10) for a dish in a restaurant
- Concrete steps: What is average weight of a bag of chips? We know $\sigma = 2$ gms. We sample 10 bags and find that they weigh 99,100,102,101,100,101,100,99,100,101
- We first find $\bar{x} = 100.3$, then we can say that the 95% confidence interval around μ is $(\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 100.3 \pm 1.96 \frac{2}{\sqrt{10}} = \{99.06 \leq \mu \leq 101.54\})$
- One sided bound?



Other confidence intervals



- When variance is unknown: $\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$
- Examples: Same as when variance was known
- Large-sample confidence interval for proportions:
- Examples: Sales calls
- Confidence interval on the variance:

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\frac{(n-1)s^2}{\chi_{\frac{\alpha}{2}, n-1}^2} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}$$

- Examples: Variance in the bag of chips